Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Session	**2026-27 (ODD)**	**Course Name**	**Lab : Embeded Systems**
Semester	**7**	**Course Code**	**23IOT1706**
Roll No	**61**	**Name of Student**	**Sheikh Wasimuddin**

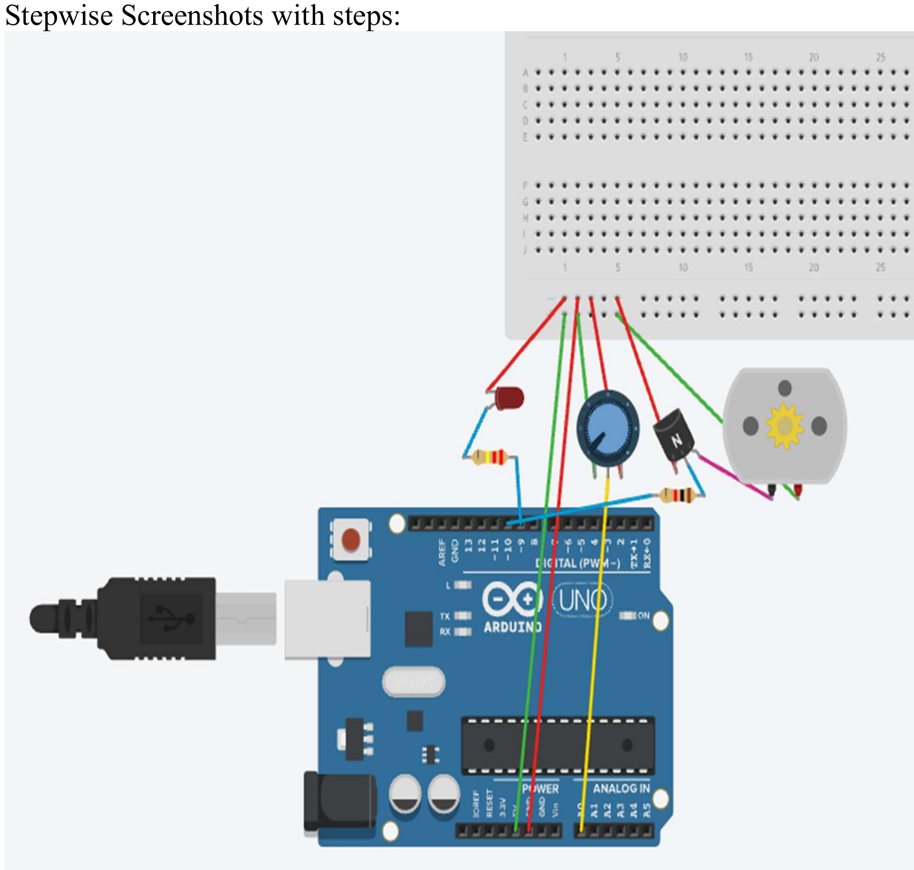
Practical Number	**06**
Course Outcome	**CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools.** **CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals.** **CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols.** **CO4: Design, simulate, and test embedded system applications for real-world applications.**
Aim	To generate a PWM signal using Arduino Uno and control the brightness of an LED and the speed of a DC motor by varying the PWM duty cycle using a potentiometer in the Wokwi simulator.
Theory (100 words)	**Pulse Width Modulation (PWM)** is a technique used in embedded systems to control the average power delivered to electronic devices by rapidly switching a digital signal between **HIGH and LOW** states. The amount of time for which the signal remains HIGH in one complete cycle is called the **duty cycle**. Arduino Uno provides PWM output on digital pins marked with the ~ **symbol**. The analogWrite() function is used to generate PWM signals, with values ranging from **0 to 255**. A value of 0 represents 0% duty cycle, while 255 represents approximately 100% duty cycle. In this experiment, a **10 kΩ potentiometer** is used as an analog input device. Its output is connected to the Arduino's **A0 pin**. The analogRead() function reads the potentiometer value in the range of **0 to 1023**. This value is then mapped to a PWM range of **0 to 255**. The PWM output is applied to an **LED** and a **DC motor**. The brightness of the LED changes according to the PWM duty cycle. For controlling the DC motor, an **NPN transistor** is used as a switching device. The PWM signal controls the transistor, which varies the average power supplied to the

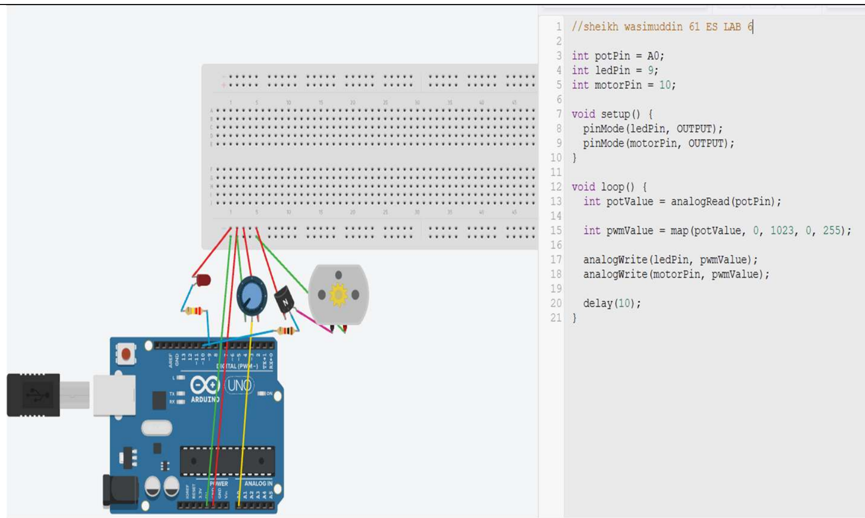
	motor and hence changes its speed. When the potentiometer is rotated, the analog input value changes. Arduino converts this value into a corresponding PWM value, allowing the **LED brightness and motor speed to be controlled simultaneously**.
Procedure and Execution (100rds)	1. Open Tinkercad Circuits and create a new circuit. 2. Add an Arduino Uno R3 and a breadboard. 3. Add the required components: ➢ 10 kΩ potentiometer ➢ LED ➢ 220 Ω resistor ➢ DC motor ➢ NPN transistor ➢ 1 kΩ resistor 4. Connect the Arduino 5V pin to the positive rail of the breadboard. 5. Connect the Arduino GND pin to the negative rail of the breadboard. 6. Connect the potentiometer: ➢ One outer pin → 5V ➢ Middle pin → A0 ➢ Other outer pin → GND 7. Connect the LED: ➢ Arduino D9 → 220 Ω resistor → LED anode ➢ LED cathode → GND 8. Connect the DC motor through the NPN transistor: ➢ Motor positive → 5V ➢ Motor negative → Collector (C) ➢ Transistor Emitter (E) → GND ➢ Arduino D10 → 1 kΩ resistor → Transistor Base (B) 9. Enter the PWM program in the Arduino code editor. 10. Start the simulation in Tinkercad. 11. Rotate the potentiometer and observe the change in its analog value. 12. Observe that the LED brightness increases or decreases

according to the potentiometer position.

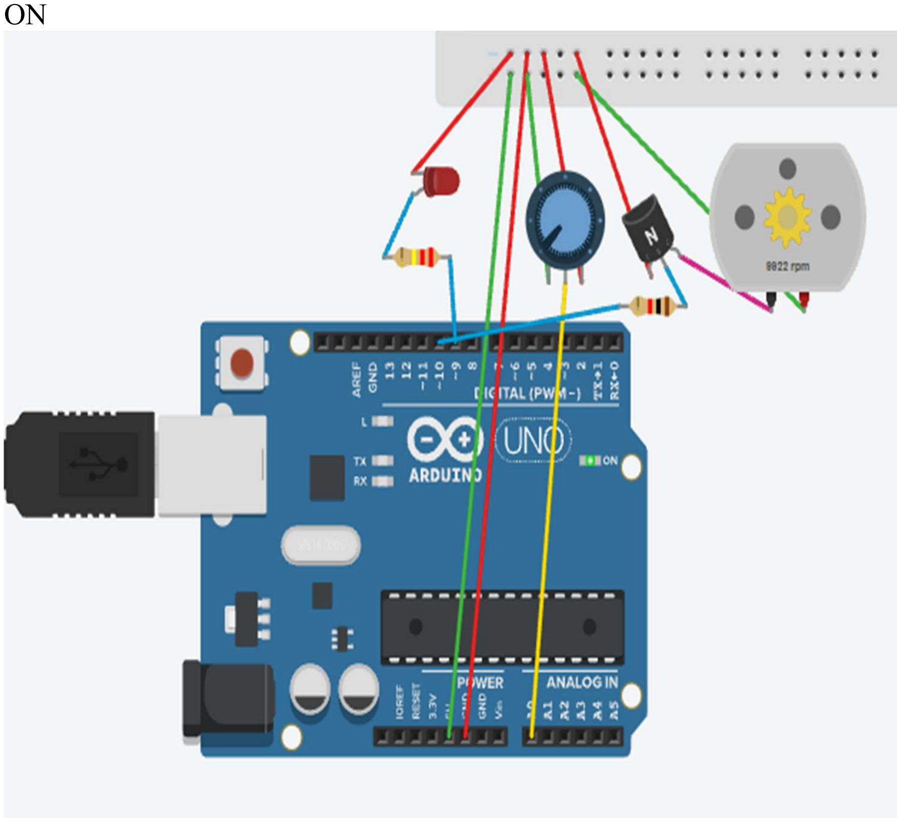
13. Observe that the DC motor speed changes along with the PWM output.
14. Rotate the potentiometer to different positions and verify the corresponding changes in LED brightness and motor speed.
15. Confirm that the potentiometer successfully controls both devices using PWM.

Stepwise Screenshots with steps:




//sheikh wasimuddin 61 ES LAB 6
int potPin = A0;
int ledPin = 9;
int motorPin = 10;
void setup() {
pinMode(ledPin, OUTPUT);
pinMode(motorPin, OUTPUT);
}
void loop() {
int potValue = analogRead(potPin);
int pwmValue = map(potValue, 0, 1023, 0, 255);
analogWrite(ledPin, pwmValue);
analogWrite(motorPin, pwmValue);
delay(10);
}

ON


9922 rpm
ARDUINO
UNO
ON
POWER
ANALOG IN

Conclusion	The PWM experiment was successfully performed using **Arduino Uno, a potentiometer, LED, and DC motor**. The potentiometer was used to provide an analog input, which was converted into a PWM value. The PWM signal controlled the **brightness of the LED** and the **speed of the DC motor** through an NPN transistor. It was observed that increasing the potentiometer value increased the PWM duty cycle, resulting in higher LED brightness and motor speed. Thus, the experiment successfully demonstrated the practical application of **PWM for controlling electronic devices using Arduino Uno**.
Date	20-08-2026